

Taeyun Roh

Seoul, Republic of Korea | txnrd.github.io | github.com/txnrd

AI Researcher building autonomous agents powered by advanced visual reasoning to thoroughly understand their environments.

Education

Korea University <i>M.S. in Computer Science and Engineering</i>	Mar. 2026 – Present Seoul, Republic of Korea
Korea University <i>B.S. in Computer Science and Engineering (GPA: 4.08 / 4.5)</i>	Mar. 2022 – Feb. 2026 Seoul, Republic of Korea

Research Experience

Data Mining & Information Systems Laboratory, Korea University <i>Undergraduate Researcher, advised by Prof. Jae Woo Kang</i> Research on agentic memory systems for AI agents.	Mar. 2025 – Feb. 2026 Seoul, Republic of Korea
Network and Mobile System Laboratory, KAIST <i>Research Intern, advised by Prof. Sung-Ju Lee</i> Worked on efficient LLM inference, including on-device LLM inference.	Jan. 2025 – Mar. 2025 Daejeon, Republic of Korea
System Software Laboratory, Korea University <i>Undergraduate Researcher, advised by Prof. Gyeongsik Yang</i> Researched federated learning for healthcare applications under resource heterogeneity.	Jan. 2024 – Dec. 2024 Seoul, Republic of Korea

Selected Publications

Distilling Expert-level Planning for Real-world Diabetes Prescribing Yein Park, Sihyeon Park, <u>Taeyun Roh</u> , Mujeen Sung, Jaehoon Yun, Sooyeon Lee, Sanghoon Lee, Jaewoo Kang ICML 2026 Workshop on Structured Data for Health	
CLAG: Adaptive Memory Organization via Agent-Driven Clustering for Small Language Model Agents <u>Taeyun Roh</u> , Wonjune Jang, Junha Jung, Jaewoo Kang	[Project Page] ACL 2026
Breaking Failure Cascades: Step-Aware Reinforcement Learning for Medical Multimodal Reasoning Junha Jung, Minbyul Jeong, Suhyeon Lim, Sungwook Jung, Jaehoon Yun, <u>Taeyun Roh</u> , Mujeen Sung, Jaewoo Kang	Under Review
When Choices Become Priors: Contrastive Decoding for Scientific Figure Multiple-Choice QA <u>Taeyun Roh</u> , Eun-yeong Jo, Wonjune Jang, Jaewoo Kang	[Project Page] Under Review

Industry & Development Experience

Aigensciences <i>AI Researcher</i> - Built an LLM-based pipeline for automated hypothesis generation in drug discovery. - Developed a pediatric counseling chatbot designed for KakaoTalk integration.	Mar. 2025 – Aug. 2025 Seoul, Republic of Korea
Blurting <i>Frontend Developer, Co-founder</i> - Co-founded a four-member startup and led end-to-end frontend development. - Developed and operated a blind-dating application that reached #73 in the Korea App Store Social Networking category.	Jun. 2023 – Jan. 2025 Seoul, Republic of Korea

- Built mathematical models for optimizing teaching assistant compensation.
- Applied regression analysis to study the relationship between student study time and academic performance.

Awards

- **1st Place, Korea University Capstone Design** (Prize: KRW 1,800,000) — Developed *NEWSQRAB*, an AI-driven system for automated news-to-short-form video generation using RAG-based LLMs.
- **2nd Place, Korea University NEXT Software Competition** (Prize: KRW 1,000,000) — Recognized as lead AI engineer for the *NEWSQRAB* project.
- **3rd Place, 2024 KIISE Conference (Undergraduate Category)** (Dec. 2024) — Awarded for the paper “*Analysis on Re-search for Data and Resource Heterogeneity in Split-Federated Learning.*”
- **2nd Place, 2023 Korea University Hackathon** (Prize: KRW 1,000,000) — Built Noonchi, a real-time visitor prediction and recommendation service for Han River parks.

Teaching Experience

Korea University
Teaching Assistant, Machine Learning (COSE 362), instructed by Prof. Jae Woo Kang

Fall 2025
Seoul, Republic of Korea

Academic Service

Conference Reviewer
ACL 2026 (ARR JAN 2026 (Secondary)), EMNLP 2026 (ARR MAY 2026 (Secondary))

Workshop Reviewer
EIML@ICML 2026

Selected Projects & Skills

Projects

- **Xalute** — Wearable health monitoring app developed with Korea University Anam Hospital for real-time health tracking and alerts.
- **Kakao Pediatric Response Chatbot** — LLM-based pediatric Q&A chatbot developed with Kakao Healthcare and Upstage.

Skills

- **Research:** Memory system for agents, Multimodal reasoning, LLM inference
- **Development:** Frontend development, applied LLM products, data analysis
- **Language:** Korean: Native, English: Fluent (TOEIC 940)

References

Prof. Jaewoo Kang
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Korea University, South Korea
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Dr. Minbyul Jeong
Principal Researcher Scientist of Large Language Model Team
Upstage, South Korea
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